

The Absence of Communism in Soviet Economic Planning

David McMullen

Melbourne, Australia

dfmcmullen@gmail.com

<https://orcid.org/0000-0001-6148-0720>

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Abstract

This paper contends that the economic malaise in the Soviet Union was mainly due to the absence of communist relations of production rather than inherent flaws in central economic planning. This is confirmed by the literature on the regime during its final decades which mostly dwelt on the behavioral failings of the three main layers of society. These layers were (1) the reactionary, oppressive and self-serving political leadership aided by an obliging bureaucracy, (2) a bonus-seeking management stratum and (3) a completely alienated, disengaged and non-revolutionary working class. The fact that the country arrived at such a sorry state is due in large part to its backward, pre-capitalist starting point. The challenge of overcoming problems such as these in the future, under more favorable revolutionary conditions, is also discussed.

In the 1930s the Soviet Union achieved remarkable results in developing the industrial base it needed to win World War II and then in the immediate post-war period to undertake a similarly remarkable reconstruction. Indeed, according to Allen (2003) the Soviet Union was the second most successful developing economy of the twentieth century, ranked just behind Japan. However, from around 1970 its performance progressively went downhill.

Those who think the problem lay in the very nature of central planning argue that it could not deal with the complexity of an increasingly developed economy trying to meet a more diverse range of requirements. This is the economic calculation argument most famously espoused by the Austrian economists (Hayek 1935). According to their view no matter how committed people were to making the system work, they would still be frustrated by its impracticality. The transformation of the relations of production and the resultant change in people's work motivation would come to naught. Everyone being perfect "socialist angels" would be to no avail.

This assessment, however, is belied by the fact that most of the literature on the economic failings of the Soviet Union point to decision making, policy and motivation problems that would not have been present if the system had been undergoing the transformation in the relations of production that you would expect of a communist revolution. Of course, central planning may still get the blame if you believe this behavioral transformation is just a utopian delusion. However, that is a different argument from the calculation one. It is simply the argument that socialist angels are not possible. This is the human nature case, not the calculation one.

The Soviet economy had a major underlying problem that I would argue was not inherent to central planning. This was the fact that most people belonged to one of three strata each of which was committed in its own particular way to obstructing production. They were: (1) a ruling stratum that presided over a police state and comprised people who were motivated by personal power and position associated with their particular domain in the system, and a desire to keep the mess functioning; (2) a management stratum that was motivated by a perverse reward system based on bonuses for production target fulfilment; and (3) a completely alienated, disengaged and non-revolutionary working class.

If the Soviet Union had been a society that was in a communist transition, you would expect to see work being transformed into something people were keen to do for its intrinsic interest and in order to contribute to the common good. Instead, work was still very much in the grip of the old oppressive division of labor that separated deciding and doing, and imposed strict top-down obedience. At the same time any attempt by the individual to help achieve good outcomes was futile. However, before we examine the pathologies of the Soviet economy and the good prospects for avoiding them in future revolutions, let us look at the essential features of the central planning system in order to provide some context.

Central Planning in a Nutshell

Under such a system, the central authorities make the big decisions on how much is spent on broad categories of final output such as new production capacity, infrastructure, the military, state machine, hospitals, schools, housing and personal consumption. Goods are allocated not through markets but by allocations through an annual plan drawn up using material balances that endeavor to ensure that the output of each product has the inputs required in its production. In other words, the aim of material balances is to ensure that the plan is internally consistent. When this meshing inevitably proved quite imperfect and input shortages start appearing the plan is adjusted with stress on ensuring the production of priority industries and goods.

The balance is constructed using norms or coefficients. A tonne of steel requires a certain quantity of inputs such as iron ore, coal, oxygen and alloys. A pair of size 10 shoes requires a certain quantity of leather, thread and metal eyelets. At the same time a balance is drawn up for labor requirements.

On the supply side you have imports, beginning-year stocks and current production available for use in the present period. On the use side you have current production, construction projects, consumer goods, exports and end-year stocks. How a product fits into these use categories varies. Shoes are not inputs into current production nor construction whereas none of the output of a steel plant goes directly to consumers. Another important consideration is production capacity. If you were planning an increase in steel production this year, you need to be satisfied that the increase in capacity is coming on stream.

The number of balances in the Soviet Union in the early 1980s was around 70,000 which indicates considerable aggregation given that the number of products was in the millions (Grossman 2008). According to Nove 1991 (p. 33) there were around 12 million identifiably different products disaggregated down to specific types of ball-bearings, designs of cloth, size of brown shoes and so on.

The accuracy of the material balance system depends very much on the information received and the ability to process it. Industries need to provide planners with their output capacities and the inputs they require for each unit of output. This has a range of uncertainties such as unexpected breakdowns, human error in the estimates, more teething problems than expected in the introduction of new technology or products, and weather conditions. Another area of uncertainty is in the completion dates for new capacity and the time required to get into full production.

Transmission of the information used to be time-consuming but with modern information technology and product codes it would now be very rapid. This would allow balances to be completed in a more timely fashion and more frequently. Also it could deal with far more products at greater levels of disaggregation both at the data gathering stage and the specification of output mix.

The process is complicated by the need to take into account indirect effects of a change in the output of a product. In drawing up a balance for a product, it is not enough to ensure the supply of the direct inputs. You also have to ensure supply of the inputs into the inputs and the inputs into the inputs of the inputs ad infinitum. So, in the case of increasing truck production, there is not just the direct increase in the use of steel, glass, rubber etc. There is also the increase in the inputs going into these inputs and so on.

Accurate balances for each product cannot be carried out independently of the others, given the competition for the same inputs. An increase in the production of one good generally has to mean less of other goods. While the balance for each product is an equation, for the balances of all goods together it is a system of simultaneous equations, in other words equations that cannot be solved separately. Given practical difficulties, Soviet planners rarely considered linkages beyond the first or second order (Dobb, 1972: 359).

In a future socialist society there is the prospect of doing far better than this. While solving a large set of simultaneous equations is impossible even with modern computers, a close approximation can be achieved in a short period of time using the Gauss-Seidel method (Cockshott and Cottrel 1993: 77 and Montias, 1959: 970). This method relies on the fact that most of the input-output coefficients are zero. For example, final consumer products such as groceries, household appliances and clothing are not inputs into anything; and sheet steel is used in many products while brake lining material has a very specific use.

With some imbalance inevitable, the next task is to ensure that industries or products receive inputs in short supply according to a priority ranking. In the 1930s and 1940s, the priority was quite correctly war production and then reconstruction. In the 1950s and 1960s living standards rose significantly, but military spending came to dominate.

Prices played a limited role in Soviet economic decision making. Price adjustment to ration final consumer goods was not used extensively. However, there is nothing in the nature of central planning that rules it out. It can ration stocks that have already been produced and can be a useful signal for what to produce in the future.

The use of prices to choose the least cost way of producing goods was generally absent or poorly employed in the Soviet planning system. Cockshott and Cottrell (1999) argue that labor embodied would be a good cost measure once tweaked to address a number of objections raised by economists. There is still the question of using valuations to determine what should go into the plan given that resources are constrained and we cannot produce everything that we want; and likewise their use to ration capacity and inventory when imbalances arise within the planning period. No attempt will be made to resolve these questions here. Determining whether central planning can deal with all the calculation objections raised by the Austrian economists would take us beyond the purpose of this paper and the competence of the author. Cockshott and Cottrell (1993) addresses this question. (The Austrians also argue that a price system driving a decentralized form of social ownership would necessarily be a pale shadow of its capitalist equivalent. McMullen (2020) is a response to that.) Now let us look at the behavioral problems.

Everybody Behaving Badly

If the Soviet Union had been pursuing communism, people would have been behaving like revolutionaries but that was not the case. This was true at every level: the leadership, the enterprise managers and the rank and file.

The political system was a thoroughly opaque police state where economic decisions served the interests of the political and the executive elite. The political elite ranged from the Central Committee down to first secretaries of provinces, regions, and cities while the executive elite ranged from members of the Supreme Soviet and Council of Ministers, to minister presidents of the republics and chairmen of regional and city executive committees, down to the directors of the hundred or so largest enterprises. Fewer than a thousand individuals belonged to this select group. (Gregory 1990, p. 2)

Ministries and local party bosses were not interested in the big picture but rather in the size and performance of their own domain. Those leading the military and heavy industry sectors had particularly strong influence. Harrison (2003: 25) suggests that military spending may have been around one fifth of GDP in the 1980s. The competing claims led to the Council of Ministers coming up with impossible production targets. However, planners did not see it as their job to object, and would not dare (Gregory 1990: 34). Instead, to comply, they drew up plans on the basis of very "brave" assumptions that inevitably lead to economic disruption.

At the middle level of the economy you had enterprise management. The big problem here was the incentive system, accompanied by excessively taut production targets. For enterprise management personnel a significant proportion of income took the form of bonuses. These payments plus career prospects depended on reaching production targets. These incentives had perverse effects that were supercharged in the presence of taut planning. This required plants to work at full capacity with little slack in the system. Insufficient output was assigned to inventory. This was the case both with direct inventory, and the indirect form where there was both inventory and idle production capacity upstream that would be used once it was clearer what specific product assortment was required. To achieve the target, management had to not only ensure everything ran smoothly with little room for slippage but to also contend with the input delays and shortages that were bound to be made more severe with taut planning.

One perverse effect of these incentives was on the information flow required by the system. Planners were dependent on management to provide them with the information they needed in order to come up with their material balances and the resulting production targets. One should not be surprised that management tried to get a soft plan by understating their production capacity and overstating their input requirements.

Supply problems were made worse by hoarding on the part of ministries and enterprises. The opportunity to do this occurred for a number of reasons. Inputs may be unusable because a complementary one was in short supply. Constant changes in the plan meant it was not unusual to have stuff you did not need as well as a shortage of things you did need. There were also supply errors. If these hoards did not turn out to be of future use, they were handy for under the table barter which was a major feature of the system.

There was also labor hoarding. It paid managers to retain workers so that they would be available when needed, particularly under conditions of labor shortage. At the same time they generally did not have strong incentives to minimize their labor costs. Indeed, overmaning occurred on a vast scale. Figures of 15-20 percent were bandied about (Ticktin, 1992, p.141). As well as labor shortages and hoarding, there was also regional unemployment because of restrictions on mobility (Ticktin, 1973, p. 31).

Investment hunger was another cause of supply difficulties as enterprises pushed for increased capacity to ease the task of meeting their targets. This was aggravated by the ratchet effect where “planning” tended to mean instructing enterprises to produce what they did the previous year plus an extra few percent (Roland 2008: 5). At the same time there was little reward for being economical in the use of capital; and, as we will discuss below, the incentive system discouraged innovation as a way of increasing output. If fed selected data, project evaluation techniques could be relied on to give the desired answer (Eatwell et al: 98). Supply constraints meant construction delays and diversion of materials away from final consumption including housing, retail outlets, urban transit and factories producing consumer goods. Consumers had no leverage while ministries producing upstream producer goods did. All up, Soviet estimates suggested that in the mid 1960s 25 percent of all working time in the USSR was being lost through difficulties with the supply system (Ellman 2014: 37).

Achieving 100 percent plan fulfillment was critical for management. If you reported 99 percent fulfillment, you lost your bonus. This led to cutting corners, particularly as the deadline approached. This took many forms, and was always at the cost of users.

With supplies running late, there would be “storming” to get output completed in time. In the mad rush, quality suffered and equipment was not properly maintained. You had shaving on quality, particularly in ways that were not easily detected. Examples included fewer stitches in a garment, fewer screws in a component, greener lumber in the building and more impurities in metal (Douglas and Patman 1960: 359).

The extent of poor quality was remarkable. According to one report, dental fillings only lasted a few months because of the poor quality of the material (Nove, 2011, p. 77). The breakdown of engineering machinery was estimated to be three to four times as frequent as in the U.S. Because

of this high breakdown rate, the Soviet Union had more people repairing machinery than actually making it (Ticktin 1973: 25). The breakdown figures for tractors were startling. The Soviet Union produced far more tractors than the United States but had far fewer per hectare in service. Their tractors broke down sooner and more frequently, and repairs cost far more than production (Ticktin, 1992: 124).

According to 1986 figures, a third of the cost of refrigerators went on repairs. The would have been higher if all returns were accepted and consumers were more demanding (Ticktin 1992: 98). Tires and televisions were on a long list of goods that wore out more quickly than might be expected in the current state of knowledge and technique (Nove 2011: 78). Vast quantities of goods were rejected by distributors because of poor quality (Chi 1978:41) while, in spite of general shortage, retailers found themselves stuck with unsaleable stock (Ticktin 1992: 134).

In the presence of shortages, construction enterprises happily made substitutions that were quite unsuitable for the customer but nevertheless allowed them to meet completion deadlines. Low-grade cement was a classic example. (Gregory 1990: 118)

One of the most serious effects of focusing on meeting production targets was management tardiness when it came to innovations in products or production methods because their introduction was bound to temporarily disrupt production through teething problems, retooling and having to learn new tricks (Douglas and Patman, 1960: 364 and Dobb 1970: 36).

Famously, the primary target indicators used to determine bonuses also distorted output in numerous undesirable ways. As a result, goods were not only shoddy and late, they were not what was required. There were a diverse range of indicators, all with their own particular perversities. They included weight, length, gross value, value added and profit.

Quantifying the production target was a tricky business. Output can have various dimensions. They can have different sizes and thicknesses, and be adapted to particular uses and requirements. Customers would require a certain assortment. The trouble was that one measure tended to take precedence when assessing target fulfillment. When push came to shove, and the deadline was approaching, the other features tended to miss out. The features that most easily met the target tended to be chosen.

If the target was in weight, large or heavy items might get you to the target more readily. Larger and heavier nails, chandeliers and bedsteads are among the proverbial examples. Nove (1958: 5) cites the case of a manufacturer of roofing iron which increased its output in tons by 20 percent but only by 10 percent in square meters. Dobb (1970: 34) tells us about spinning mills producing thick yarn of low count, excessively thick glass and paper, and machinery that was typically 20 percent heavier than its western counterpart. Measurement in length could have similarly horrific

consequences (Dobb 1970: 34). Pieces of cloth would be narrower than required and the easiest weave.

Where output is heterogeneous rather than uniform and standardized, the easiest measure was in terms of gross value where items were added-up at their current selling-price. In this case expensive materials were used. There was marked reluctance to use any but the dearest kinds of steel in tool-making. Clothing factories produced more than the required number of garments made from the best cloth (Dobb 1970: 34). The "gross value of construction" target gave construction enterprises the incentive to substitute more expensive materials and more expensive labor, even when this was not required by the project's technical design (Gregory 1990: 118). Perhaps the oddest effect of using gross value was the effect on the supply of spare parts. Take the example of a spark plug for a tractor. If it is supplied separately it is only credited with its own gross value whereas if it is incorporated in a complete tractor it is part of a product the gross value of which contains numerous other components purchased elsewhere. The shortage was compounded by the fact that shoddy products greatly increased the need for more spare parts (Dobb 1970: 35).

If the output indicator was value added, there was an incentive to be extravagant in the use of labor and to maximize the amount of work carried out within the enterprise rather than acquiring it from other firms which may allow for greater specialization and economies of scale (Nove 1958: 6).

If profit was the indicator they would use the cheapest materials, spend the least possible time and choose products with the highest prices. This might seem fine at first glance except that it tended to mean poorest quality materials, shoddy work and producing the goods that are high priced for reasons that had nothing to do with high demand, including government efforts to discourage consumption. (Ticktin, 1973, p. 33)

Another effect of the supply problem was inefficient vertical integration. Enterprises often had to produce in-house to get around the problem of unreliable suppliers. It is critical for components, sub-assemblies, spare parts, ancillary equipment, etc to meet certain quality standards and exact specifications. They also need to arrive in a timely fashion if production is not to be disrupted (Hunter 1961: 567). An added reason for in-house production was the difficulty in getting outside producers to make specialized machinery. Enterprises preferred mass produced goods that allowed them to be more "flexible" in what they supplied (Gregory 1990: 64). It was quite common for enterprises to have a small casting or foundry shop (CIA 1956: 51) and do their own construction work (Gregory 1990: 63). Vertical integration also occurred at the ministry, glavk and industry association level, and included raw materials extraction and transport. It was considered more reliable than dealing with "foreign" organizations. Vertical integration added to

costs in three ways: makeshift methods, failure to exploit economies of scale and management having to supervise an excessively large and diverse number of activities and technologies.

All this underhand behavior of managers was greatly assisted by so-called family circles who would cover for each other (Gregory 1990: 80). These included superiors and subordinates who benefited from enterprises having good performance numbers, and besides, superiors would want to avoid scandals that they could be held responsible for. You also had old school friends and receivers of past favors or bribes. Not being a "team player" could have dire consequences. For example, Douglas and Patman (1960: 376) refer to the case of a naïve young chemist who showed that a production sample had too much phosphorus and found himself fired on trumped-up charges. People were shuffled around in an attempt to counter the family circle problem. However, this caused organizational disruption and a short term view among those who expected to be shuffled. Perverse behavior was also assisted by the need to "maintain good relations". You did not complain about your suppliers because they could retaliate and make things hard for you in the future (Gregory 1990: 68).

Now we come to the rank and file workers. Their alienated state was another major source of poor economic performance. There was a total absence of the intrinsic motivation that is meant to emerge in a society undergoing a communist revolution. As with management, incentives were of the extrinsic variety. Piece rates were king. Because of the principal-agent problem this is an inferior outcome to one where workers are interested in their work, are developing their abilities and wish to contribute to the common prosperity. Furthermore, the Soviet Union was quite weak even with this inferior form of motivation. Generally there was not the threat of unemployment to discipline workers and wages were no inducement because there were so few goods available to buy.

Workers did not have to worry about unemployment. The regime went to great lengths to avoid abolishing jobs from under people. They were rather tardy about closing down obsolete or excess plants, and people were not made redundant with new technology. You were only sacked in the case of major transgressions. Job security was a political imperative. It was seen as a core feature of socialism and much was made of how unemployment was the hallmark of capitalism. Adding to the political difficulty of shedding labor, was the fact that there was often a considerable range of benefits associated with your present employment, including housing, childcare, cafeterias and medical facilities. This was part of attracting and keeping workers when the state was failing to supply final consumption goods and services. At the same time firms were keen to hoard labor because of the difficulty of getting it when needed and the fact that there was generally no benefit to them of controlling their wage bill. If you left a job there were plenty of others available because of the general state of labor shortage created by the system.

Higher pay was no great inducement because of the shortage of consumer goods and the fact that many goods and services were free or available at very low prices. "We pretend to work and they pretend to pay us" is a well-known phrase from that time. What you got was "socialist shirking". One Soviet commentator claimed that only a third of workers worked properly (Ticktin, 1992: 102).

Management had no strong incentive to get the most out of workers and were content with a slack pace as long as it was predictable. The effective powers of foremen were weak. They were not backed up from above and did not want to seem to be a management stooge. (Ticktin, 1992: 97).

Excessive alcohol consumption was an increasing problem (Anderson and Hibbs 1992: 441). Labour productivity after a weekend fell significantly as a consequence of drunkenness; and it was also the cause of many accidents at work. Between 1940 and 1980 consumption increased by 600 percent while the population only increased by 25 percent. By 1980, the average Soviet family was spending 25-50 percent of its monthly food budget on alcohol.

Revolution would have been the Solution

If the Soviet Union had been undertaking a communist revolution, it would have dispensed with these very crude forms of material incentives. While rewards for the quality of work would have been unavoidable for some time, you would expect them to play a diminishing role. A critical part of that on-going revolution is the creation of totally new relations of production; and in such a process, work increasingly becomes something people want to do for its own sake rather than for reward. Intrinsic motivation takes over from material or extrinsic incentives. So bonuses and piece rates become a thing of the past. This is achieved through a transformation in the nature of work, and in our relations with others, both at work and in society at large.

Internal or intrinsic motivation means that people find the work itself rewarding and want to contribute their bit to the common pot from which they all share. Pay for performance would become increasingly less important as the new conditions develop. One can also possibly imagine progressively less need to financially induce people to work longer than they would prefer or to take on a job that is not their first choice. The knowledge that they are contributing to a bigger and better common pot would be sufficient. The change in motivation would eliminate the principal-agent problem because we all become principals. We want to contribute to good outcomes because we are direct beneficiaries, and we are not presented with perverse incentives to do otherwise.

This transformation in the nature of work and our attitude to it would require on the one hand the increasing reduction in the more tedious forms of work and on the other an ongoing revolution

that engages the vast majority in the struggle to transform our relations with one another in production and in society generally. Later we will discuss the nature of this revolution and how it was aborted in the Soviet Union, but firstly let us look at how it would have made all the difference to the Soviet planning system

Planning under Communist Conditions

If you assume the communist transformation in the relations of production, the problems we have discussed would not exist because people would behave in the opposite fashion. Let us start by looking at hierarchy and subordination. The Soviet Union was characterized by military style top-down obedience and no freedom to speak your mind. However, under increasingly communist conditions you would see a general reliance on mutual agreement between different levels and pushback against poor decisions from above. Assessment and supervision would also go both ways. Furthermore, there would be low tolerance for pathologies such as graft, patronage, regionalism, departmentalism and empire building; and at the same time there would be less desire to engage in such behavior and no inducement to form family circles.

Ministries and enterprises would have more say over their production targets and supply allotments. They would refuse to accept arbitrary and unrealistic plans. At the same time these organizations would be full of assiduous people keen to maximize outputs and minimize inputs.

Enterprises would allow sufficient downtime to ensure there was adequate repair, maintenance and replacement. They would also endeavor to make good trade offs between innovation and the disruption of present production. Coordination with customers and other suppliers would minimize problems in both of these situations. Care for customers would be a core value whether liaising with them directly or through distribution organizations. This would mean better timing, assortment and quality, and willingness to take on special orders.

As well as better management decisions, we would also see a much better approach to work by the individual worker. The problem of "socialist shirking" would recede. Everyone would ensure the quality of their own work and show low tolerance for poor upstream work that badly affected their own. Likewise, as consumers, they would not tolerate poor quality of consumer goods and services, and inadequate provision of wholesale and retail facilities.

Everyone would be constantly engaged in innovation. There would be better integration of research, development and deployment work. The elimination of old jobs and the creation of new ones would be a very common occurrence. No one would want the same old job for life. People would be increasingly versatile and equipped to be lifetime learners. They would be constantly studying to take on new work either where they are or elsewhere. In the adoption of new technologies there would be stress placed on eliminating the less congenial work. Workers would

not allow themselves to be hoarded. If you are not needed at the moment where you are, you would expect to be assigned elsewhere or take on training and study to prepare you for new work.

A First Go under the Worst Conditions

So, what we have just discussed would be what we would have expected if the Soviet Union had been taking the road of revolutionary transformation. However, that trajectory was necessarily going to be especially difficult and protracted in the Soviet Union because of its extremely backward starting point; and then eventually political developments presented a brick wall.

In 1917, the Bolsheviks found themselves with none of the things you would want for a proletarian revolution. Proletarians (i.e., wage workers) were a small minority. Something like 80 percent of the population were peasants who were only concerned with seizing land from the landlords. Most of them had never heard of the proletarian revolution and were not really interested when they did. The Bolshevik victory in the 1917-1921 civil war owed a lot to the fact that they supported the peasants' demands while their opponents did not. They made their bid for power in the hope of similar revolutions in Europe. The idea was that the more advanced region would assist the more backward Russia while the power seizure in Russia would spur events in the west and prevent the country from supporting counter-revolutionary forces there.

The basis of communist relations were a long way off. In agrarian pre-industrial societies classes and inequality were inevitable. Equality would have meant sharing poverty and toil, and so everyone was keen to get more than others and do less of the work. Everything was unavoidably nasty, and it was normal to have kings, lords and rampaging armies.

Only with developed capitalism do we break with these underlying conditions. Living like everyone else is no longer wretched and life threatening. And a typical day's work no longer has to be long, backbreaking and totally mindless. Besides, an average share of the pie would not leave you missing out on much, with improvements in material conditions coming from a bigger and better pie. There is also less benefit in avoiding work as it becomes less tedious and routine and there is a diminishing need for a division of labor between thinking and doing, and more the all round exercise of our human capacities.

So we achieve conditions where the benefits of mutual regard outweigh any benefits from a bigger share and the avoidance of a standard day's work. What we would gain from being a "winner" in a class society no longer makes up for the lack of general good fellowship. By having regard for one another instead of abusing or being indifferent to them we gain a great deal. When we help others to be fully functional, interesting, happy and hopeful they in turn

enrich our lives. People are not of much benefit to us if they are boring, disturbed or bitter and twisted, indeed they are often quite harmful.

In the initial period in the Soviet Union, backwardness was the binding constraint that held the revolution back. However, this changed in the 1950s once the economy had the ability to deliver the beginnings of material prosperity. The binding constraint then became the political line and practice. This relegated the revolution to a distant future.

The Soviet leadership subscribed to what has been dubbed the theory of the productive forces. This was couched in terms of the primary contradiction in socialist society being between the advanced social system (socialism) and the backward productive forces. You already had the new social system so all you needed now was economic development. Reliance on material gain and the old relations of production would remain until there was overabundance and no longer a difference between mental and manual labor.

However, the reality proved that if you do not make significant changes in work motivation well before some misty communist future, you end up with the mess we witnessed. At the same time you find yourself with a proto bourgeoisie among the cadres who game the existing system while being increasingly lured by the siren song of "market solutions"; and complementing this is a politically backward and subdued working class.

It is also worth noting here that when you look at their economic failings, they were the greatest in those areas that would have assisted the move in a communist direction. They failed to make equality palatable by devoting sufficient resources to housing, public transport, consumption goods and health care. The relations of production ensured that agriculture and farm produce distribution were a particular mess. The privileged layers of society, however, did not have to worry because they had special access. The failure to innovate meant that far too many jobs were particularly humdrum. The fetch and carry, and machine tending variety that were being eliminated in the West still predominated.

The Revolution Next Time

What happened in the Soviet Union is in marked contrast with what we could expect in a future revolution under conditions of advanced capitalism. Then the prospects will be far brighter. To start with, as we have just mentioned, conditions in a modern industrial society are far more conducive. You have a level of productivity where an equal share would be shared prosperity rather than shared poverty. Then you have advances in technology that give an increasing proportion of work the potential to be a desirable activity, while sharing the diminishing amount of less desirable work would not be too onerous. At the same time, it would be a full blown proletarian revolution and not just a group of communists piggy backing on a peasant revolution.

It would only occur in the first place if a large proportion of the “90 percent” were committed to the project and formed themselves into a powerful revolutionary movement. Then the true grit displayed in seizing and establishing a revolutionary government, with all its *sturm und drang*, would carry over into the period when we transform ourselves and society. We would not want to leave the job half done and then see it overturned. In this revolutionary struggle we will have to overcome the shortcomings we inherited from capitalism as well as the resistance from supporters of the old society.

So, armed with the more favorable material conditions, the revolutionary movement would among other things set about transforming work into a place where people thrive, want to be and want to contribute. This will require, on the one hand, eliminating the old, and ever less technologically necessary, division of labor, and on the other hand, changing how we generally treat each other in the workplace.

The division of labor under capitalism tends excessively to separate thinking and deciding from doing. Work is oppressive and narrow. This makes sense under capitalism where unwilling wage slaves have to be supervised and told what to do, and a hierarchical career structure is needed to give a middle stratum an investment in the system. It will not be possible to eliminate these old ways of doing things overnight. It will take time for the rank-and-file to take on their wider role. They will have to get an education worthy of the name, struggle up all kinds of learning curves, break down fears and gain confidence. At the same time they will have to push back against people who want to slow down the process. There will be resistance and foot dragging from various quarters. These include die-hard opponents, the old management that cannot be immediately dispensed with, people who are simply comfortable with how things are and find change too messy, and supposed revolutionaries who simply want to take over the old positions and not really change anything.

Personal relations between people will have to be put on a more human footing if work is to be something we are generally enthused about, and become a place where we thrive and develop. We generally have a lot to learn when it comes to developing productive and gratifying relations with others. There is a lack of social skills and empathy. We are not that good at clearing up misunderstandings or resolving conflicts. We are not always helpful. Then there are people who have specific behavioral difficulties, often with clinical names. They have anxieties and other disorders that cause their interactions to be disturbing for themselves and others. They are not having a good time and they are no fun to be with. Next there are “bad actors” who engage in bullying, scheming and lying, and are in many cases sociopaths or psychopaths. Often they will be conscious opponents of the revolution.

We will need a whole range of new capacities if we are to help those who need help and push back against those who behave badly. These include greater self-confidence, emotional

intelligence and diverse new social skills. Another challenge will be casting off passive, submissive and weak-spirited habits engendered by our subordination under capitalism.

Fortitude and commitment will be required to deal with bad situations when it would be easier to keep quiet or find another job. Moral courage will be another must-have when dealing with bad actors because they tend to retaliate in devious ways. They may set you up, claim that you are the problem and not them, take advantage of your mistakes and use any position of authority they may have. Dealing with people in authority or of high standing who have started to be a hindrance rather than a help will be particularly tricky. There may be a view or practice that is widely accepted but you think it is holding things back (an "ill wind"). When you raise your objections there may well be a nasty backlash from those who have a lot invested in what you are challenging.

This arduous battle for change is the classic collective action problem. Collective action requires you to do your bit even when the pain of doing so is greater than any direct personal gain. However, in doing so you are contributing to a change in culture that you and those close to you will benefit from at other times and places.

This is often referred to as the free rider problem. There is the temptation to slacken our efforts because our lesser individual contribution will be unnoticeable. However, if a lot of people act that way it becomes very noticeable. Once this happens the more conscientious will see no sense in going out of their way to contribute to good outcomes. Their efforts will come to nothing and so they give up

This is where the revolutionary movement has to come in, particularly the more committed members. The success of the revolution is extremely important to them all. They are determined to play their part, despite the storm and stress, and know that their comrades are the same. With this initial critical mass, people at large can see things happening and that it is worth contributing. They want to contribute as they see a society unfolding that is increasingly meeting their needs.

We will create a society where problematic behavior activates counteracting responses that act as a negative feedback mechanism, a thermostat which creates a virtuous circle that leads to progressively better behavior over time. This is different from what we have at the moment where bad behavior tends to be met with passivity and encourages more bad behavior, where bad behavior drives out good behavior. Morality and conscience will also be part of the picture. Not only will you be deterred from doing the wrong thing by the response of others, you will feel bad about it.

Conclusion

We have shown that most of the performance failures of Soviet central planning were not inherent to such a form of organization but rather were due to the absence of a vital ingredient, namely, emerging communist relations of production. Just as capitalism requires its own relations of production so too does the society that supersedes it. The shift from one to the other is from unwilling to willing labor, and from dog-eat-dog to other-regarding behavior. When explaining the difference this quote from the *Communist Manifesto* is hard to beat:

In the place of the old bourgeois society with its classes and its class antagonisms, there will be an association in which the free development of each is the condition for the free development of all.

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